HABITS OF HIGHLY EFFECTIVE
ap psychology learners

Name	Praveen	Period	1

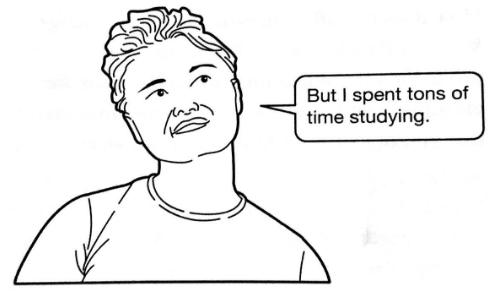
Directions: Read the following excerpts about effective learning and answer the questions at the end of the reading. You should keep this returned assignment in your binder for the duration of the school year to revisit important learning principles.

The Advanced Placement Psychology course aims to provide students with a learning experience equivalent to that of most college introductory psychology courses. In order to earn college credit students must earn a score of "3" or higher on the AP Psychology exam in May. It is recommended that all students strive for a "5" (the highest score) on the exam. Getting a "5" will require highly effective studying and learning practices. Implementing the tips in this reading into your regular study routine will help you to efficiently and effectively learn AP psychology course material.

No matter what you may set your sights on doing or becoming, if you want to be a contender, ***it's mastering the ability to learn*** that will get you in the game and keep you there. Remember that the most successful students are those who take charge of their learning and follow a simple but disciplined strategy. You may not have been taught how to do this, but you can do it, and you will likely surprise yourself with the results.

Embrace the fact that significant learning is often, or even usually, somewhat difficult. **You will experience setbacks. These are signs of effort, not of failure**. Setbacks come with striving, and striving builds expertise. Effortful learning changes your brain, making new connections, building mental models, increasing your capability. The implication of this is powerful: Your intellectual abilities lie to a large degree within your own control. Knowing that this is so makes the difficulties worth tackling. Following are keystone study strategies. Make a habit of them and structure your time so as to pursue them with regularity (Brown, 2014, pp. 200-201).

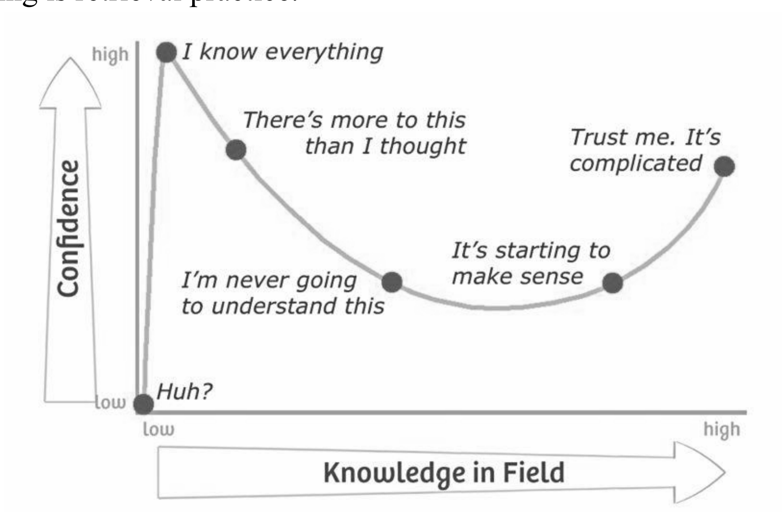
Illusion of Knowing



Can you trust what you know? It might seem like a strange question, but what I mean is, do you think you have a clear insight into what you know and what you don't know? Alternatively, are you being tricked by your brain into thinking that you know something well when actually you don't? **There are unknown unknowns-there are things we do not know we don't know.** You can only really learn effectively if you have a good awareness of what you do, and what you don't already know. Self-monitoring your learning progress and performance is critical! *You should not be finding out that you didn't actually know the material during the test*. You should find out what you know before you take the test with retrieval practice!

Dunning-Kruger Effect

Some people tend to overestimate their abilities, especially if they are still only a novice in that skill or topic. The consequence of this is that they don't see the need to improve further – they think they know it all already – and therefore, they can never improve. Interestingly, people who are already more accomplished in something don't seem to have, such as an overinflated sense of their competence, so it is an early barrier on your learning journey to watch out for. Watch out for this effect in your approach to this AP course. Something that can help avoid this bias and the illusions of knowing is retrieval practice.

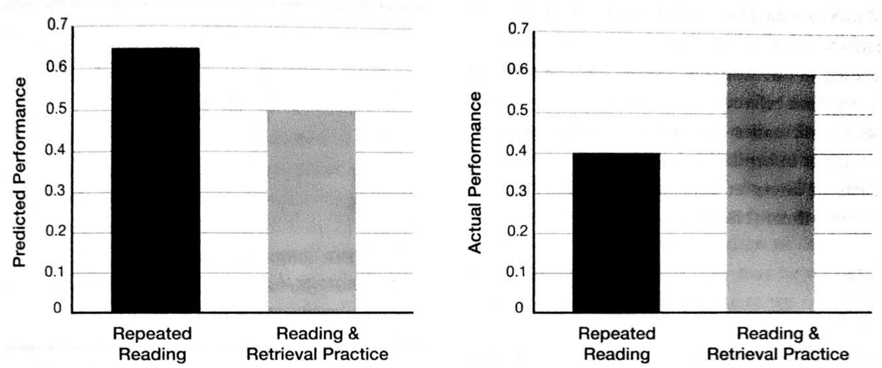


Retrieval Practice

Retrieval practice involves recreating something you've learned in the past from your memory, and thinking about it right now. In other words, a while after you've learned something by reading it in a book or hearing it in a class or from a teacher, you need to bring it to mind (or "retrieve" it). The word after is really important; you need to forget the information at least a little in order for retrieval to be effective! You don't want to just immediately recite what you see in the book or what the teacher told you, but rather you want to bring the information to mind on your own, once it starts to get a little more difficult to remember what you studied (Sumeracki, 2016).

"Retrieval practice" means self-quizzing. Retrieving knowledge and skill from memory should become your primary study strategy *in place of rereading*…

The familiarity with a text that is gained from rereading creates **illusions of knowing**, but these are not reliable indicators of mastery of the material. By contrast, quizzing yourself on the main ideas and the meanings behind the terms helps you focus on the central principles. Quizzing provides a reliable measure of what you've learned and what you haven't yet mastered. Moreover, quizzing blocks forgetting. Forgetting is human nature, but practice at recalling new learning secures it in memory and helps you recall it in the future. Periodically practicing new knowledge and skills through self-quizzing strengthens your learning of it and your ability to connect it to prior knowledge…



This graph shows predicted and actual learning after retrieval practice and re-reading. Data from Roediger & Karpicke (2006).

When you read a text or study lecture notes, pause periodically to ask yourself questions like these, without looking in the text: What are the key ideas? What terms or ideas are new to me? How would I define time? How do the ideas relate to what I already know?

If your teacher provides practice tests, or there are practice questions in your textbook, make sure to attempt them-but without looking at your book or notes! Once you are done answering the questions, make sure to check your answers for accuracy. If there are questions that you got wrong, go back to those sections in the book or your class notes and review the material. If you don't have practice questions (or you've already answered all of your practice questions a few times), you can make your own questions. This process takes a lot of time, but if you create a study group you can each create a few questions and trade.

If you're having trouble coming up with specific questions, you can just try writing out everything you can remember on a blank sheet of paper. If you have a lot of information to remember, try breaking it up into sections. You can use the headers in your textbook or general ideas provided by your teacher as prompts to help you recall as much as you can. When you are done, make sure to go back and review your class materials so that you can see what you missed and what you might need to work on more.

If you like sketching, you can try to draw everything you know about a topic from memory! It doesn't have to be pretty- it just needs to make sense to you. As long as you're drawing what you know from memory, then you're practicing retrieval.

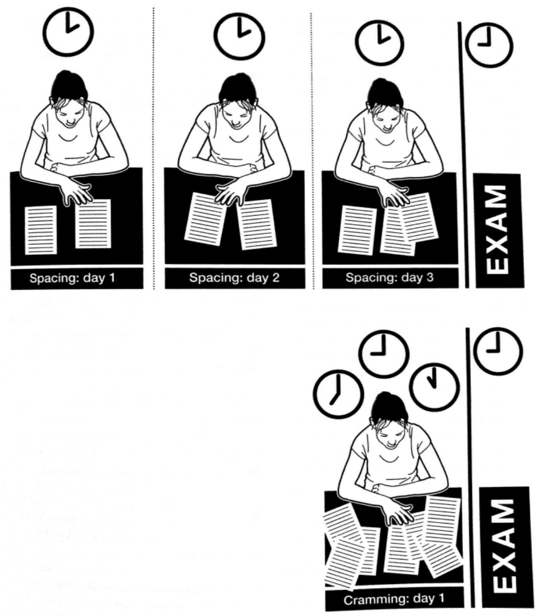
You can also create flashcards to practice retrieval. The easiest way to create flashcards is to put a question or a prompt on one side of the card, and then put the answer on the other side. To use the flashcards to practice retrieval, look at the question side of the card and try to come up with the answer. Make sure that you are really retrieving the answer. Sometimes our students say they look at the question side and have a general idea that they know the answer, but this is not the same thing as really bringing the full answer to mind. You might even consider writing the answer down on a separate sheet of paper to really make sure you're bringing it to mind. Then, after you've retrieved the answer yourself (or given it a good try) flip the card and take a look at the correct answer. There are also many apps for this (like Quizlet) if you prefer to use technology.

Set aside a little time every week throughout the semester to quiz yourself on the material in a course, both the current week's work and material covered in prior weeks. When you quiz yourself, check your answers to make sure that your judgments of what you know and don't know are accurate. Use quizzing to identify areas of weak mastery, and focus your studying to make them strong.

The harder it is for you to recall new information from memory, the greater the benefit of doing so. Making errors will not set you back, as long as you check your answers and correct your mistakes.

How it Feels: Compared to rereading, self-quizzing can feel awkward and frustrating, especially when the new learning is hard to recall. It does not feel as productive as rereading your class notes and highlighted passages of text feels. But what you don't sense when you're struggling to retrieve new learning is the fact that every time you work hard to recall a memory, you actually strengthen it. If you restudy something after failing to recall it, you actually learn it better than if you had not tried to recall it. The effort of retrieving knowledge or skills strengthens its staying power and your ability to recall it in the future (Brown, 2014, pp. 201-203).

Spaced Practice



Spaced practice is the exact opposite of cramming. When you cram, you study for a long, intense period of time close to an exam. When you space your learning, you take that same amount of study time, and spread it out across a much longer period of time. Doing it this way, t**hat same amount of study time will produce more long-lasting learning.** For example, five hours spread out over two weeks is better than the same five hours right before the exam (Weinstein, 2016).

There are at least three really big problems with cramming:

1. **First cramming actually takes more time.** Think about it: if you learn in the same amount of time spaced out (e.g., five hours in one-hour increments compared to one five-hour cram session), then you have to spend more time during the cramming session to get to the same level of learning.
2. **Second, as quickly as you learned that information, you will then also forget it.** You may do fine on the test, but all that extra time you spent during cramming? It will all have been wasted. If you had spaced your learning, you would forget much less after the test. No matter what you are learning-science, math, a foreign language, AP Psychology - future learning will depend on previous learning. ***It is therefore very inefficient to forget everything you learned for one test, only to have to re-learn it again later along with new, more complicated information!*** This also applies to future classes, where it might be helpful to retain knowledge from a previous class.
3. **Another reason why cramming is a bad idea is that it inevitably replaces sleep,** which is very important for learning and also for your mental and physical health more generally. So, resolve to form a healthy habit today and plan to space your learning!

a. Sleep is extremely important for learning. Sleep deprivation can produce a number of physical health problems such as increasing weight grain and increasing chances for illness. Sleep deprivation can also cause impairments to attention, problem solving, and decision making. What is particularly important to realize is that even mild sleep deprivation can cause these effects. Some studies show that risk to health and cognitive impairments increase if you lose 1-2 hours of sleep each night! (In other words, if you're only getting about six hours of sleep per night, your cognitive functioning, including learning, is likely to suffer.) Further research shows that getting sleep after learning improves performance later, especially for understanding information and problem solving. This is another reason that cramming (as opposed to spacing) can be so bad for your learning. When you cram, you often lose sleep the night before the exam.

Establish a schedule of self-quizzing that allows time to elapse between study sessions. How much time? It depends on the material. If you are learning a set of names and faces, you will need to review them within a few minutes of your first encounter, because these associations are forgotten quickly. New material in a text may need to be revisited within a day or so of your first encounter with it. Then, perhaps not again for several days or a week. When you are feeling more sure of your mastery of certain material, quiz yourself on it once a month. Over the course of a semester, as you quiz yourself on new material, also reach back to retrieve prior material and ask yourself how that knowledge relates to what you have subsequently learned.

If you use flashcards, don't stop quizzing yourself on the cards that you answer correctly a couple of times. Continue to shuffle them into the deck until they're well mastered. Only then set them aside-but in a pile that you revisit periodically, perhaps monthly. Anything you want to remember must be periodically recalled from memory (Brown, 2014, pp. 203-205).

Start planning early – the beginning of the semester, or even earlier. Set aside a bit of time every day, just for studying, even if your exams are months away. This may seem strange at first, if you are used to cramming right before an exam; but it's just a new habit that you will get used to if you persevere.

Review information from each class, but not immediately after class. A good way to do this is to reserve some time one day after each of your classes meet. For example, if you have classes Monday, Wednesday, and Friday, you might review the information on Tuesday, Thursday, and Saturday respectively for each of those classes.

Spacing your learning doesn't mean you won't be studying at all right before the exam. You can still study up until the exam – but instead of only studying then, spread it out so that you're studying days and weeks before the exam as well. You'll spend less time and learn more, in the short-term and in the long-term (Weinstein, 2016).

Interleaving

Interleaving-simply re-arranging the order of retrieval opportunities during spacing without changing the content to be learned-can increase (and even double) student learning. Imagine a batter practicing with a pitcher. If the batter receives 10 fastballs, followed by 10 changeups (slower pitches), and then 10 curveballs, the batter will know he only has to change his batting strategy after 10 pitches. The batter literally knows what's coming. But if the batter doesn't know which type of pitch is coming-if the pitches are mixed up-the batter will have to determine which batting strategy works best (Agarwal & Bain, 2019).

Here's what the research suggests: You shouldn't study one idea, topic, or type of problem for too long. Instead, you should change it up often. Interleaving like this may seem harder than studying one type of material for a long time, but this is actually more helpful in the long run. This strategy is particularly useful if you're studying something that involves problem solving - like math or physics - interleaving can help you choose the correct strategy to solve a problem. Interleaving can also help you to see the links, similarities, and differences between ideas (Weinstein, 2016).

Elaboration

Elaboration is the process of finding additional layers of meaning in new material. Examples include relating the material to what you already know, explaining it to somebody else in your own words, or explaining how it relates to your life outside of class.

A powerful form of elaboration is to discover a metaphor or visual image for the new material (Brown, 2014, pp. 207-208).

As you continue to elaborate on the ideas you are learning, make connections between multiple ideas-to-be-learned and explain how they work together. A good way to do this is to take two ideas and think about ways they are similar and ways they are different.

Describe how the ideas you are studying apply to your own life experiences or memories. **In addition, as you go through your day, take notice of the things happening around you and make connections to the ideas you are learning in class**. Doing this will engage an additional process that is highly effective: spacing learning over time. (Sumeracki, 2016)!

Mnemonic Devices

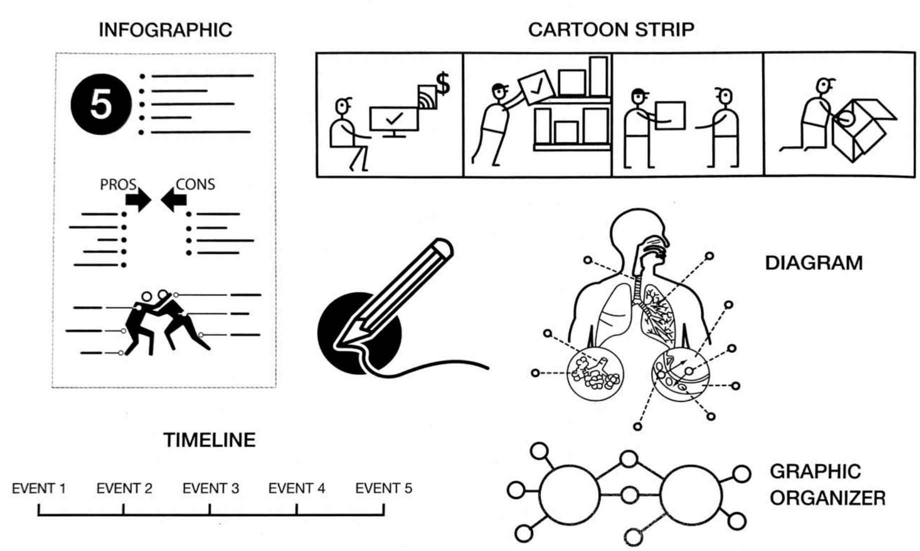
Mnemonic devices help you to retrieve what you have learned and to hold arbitrary information in memory. “Mnemonic” is from the Greek word for memory, and mnemonic devices are like file cabinets. They give you handy ways to store information and find it again when you need it.

For example, here is a very simple mnemonic device that some schoolchildren are taught for remembering the US Great Lakes in geographic order, from east to west: Old Elephants Have Musty Skin. Mnemonics are not tools for learning per se but for creating mental structures that **make it easier to retrieve what you have learned** (Brown, 2014, pp. 211).

Dual Coding

Dual coding is the process of combining verbal materials with visual materials. Pictures are often remembered better than words. Dual coding theory is the idea that when we combine text information and visual information, our learning is enhanced because we process verbal and visual information through separate channels. The idea is that when you have the same information in two formats-words and visuals- this gives you two ways of remembering the information later on. Here are some specific dual coding ideas:

- When you are looking over your class materials, find visuals that go along with the information and compare the visuals directly to the words
- Cover the text, and try to describe the visuals with words
- Another time, you can do the opposite: read the text, and try to create your own visuals
- Attempt to draw a visual representation of what you are reading in the text
- Work your way up to practicing retrieval by drawing what you know from memory



There are many ways to visually represent material, such as with infographics, timelines, cartoon strips, diagrams, and graphic organizers.

Concrete Examples

Abstract ideas can be vague and hard to grasp. Moreover, human memory is designed to remember concrete information better than abstract information. To really nail down an abstract idea, you need to solidify it in your mind. You can do this by being specific and concrete.

Take "scarcity" as an example of an abstract idea. Scarcity can be explained as follows: the rarer something is, the higher its value will be. But this description contains a lot of vague terms, such as "rarer" and "value". How can we make this idea more concrete? We could use a specific example to illustrate the idea.

Think about an airline company. If you were to try and book a flight four months in advance, the ticket prices would probably be pretty reasonable. But as it gets closer to the date of travel, there will be fewer seats left on the plane (the seats are more rare). This scarcity drives up the cost (value) of the tickets. This is a concrete example of scarcity, which is an abstract idea

So, when you're studying, try to think about how you can turn ideas you're learning into concrete examples. Making a link between the idea you're studying and a vivid, concrete example can help the lesson stick better.

Where can you find concrete examples?

- Collect examples your teacher mentions in class
- Search your books, notes, and other classroom materials for additional examples
- Look out for examples around you as you go about your day
- Share examples with friends

But be careful – not all examples you come up with are going to be equally accurate and relevant to the idea you're trying to study. Sometimes, it might be hard to tell a good example from a bad one – especially if you're fairly new to a topic. It's important to make sure your examples are correct. Ultimately, creating your own relevant examples will be the most helpful for learning; but before you get to that stage, if possible, always verify your examples with an expert (Weinstein, 2016).

Reading Questions

1. Give a specific example of the illusion of knowing in your previous school experiences. How have you been tricked by your brain into thinking that you knew something well when you actually didn't? How can you prevent the illusion of knowing moving forward?

One example of this for me was History. By taking what US-based media companies (who have a very pro-US bias) and pop history influencers (who only keep the most action-packed segments) at face value, I failed to grasp the nuance and complexity of many historical events.

To prevent this, its important to realize that almost every subject has far more complexity that it's typically initially presented with.

2. Why is retrieval practice a more effective studying strategy than rereading your class or textbook notes?

We naturally forget most of what we learn
Retrieval practice gives us several chances to reinforce the idea
It trains the brain to specifically recall info
It's scientifically shown to be more effective

3. What are your **three major takeaways** about the benefits of spacing vs. cramming?

Spaced learning is far more efficient, giving you several opportunities to relearn info after a period of time
Cramming is very short term

Also, I think neuroplasticity is a major factor. Our brains develop, and a large amount of this happens during sleep. By spacing a, for example, 3 hour study session over 3 days, not only do you get 3x the opportunity to develop the neural pathways, but the 2nd session will be more productive because not only do you have 1 hour of knowledge on the subject, you also have more knowledge / understanding from the brain development. This effect would continue, and I predict that it would have an exponential effect compared to a single cram session

4. Give an example of how you will use the following learning strategies in our course:

a. Interleaving
Switch between various subtopics, or for me, switch between Psych & Calc

b. Elaboration
Since Psychology is a very human science, I can find ways a subject relates/applies to me
Also, I've found seeing a deeper meaning by doing my own research on a subject can help better understand it by seeing it in new contexts

c. Mnemonic Devices
Especially for structured lists of info, e.g. parts of the brain, it would be trivial to come up with a Mnemonic device.

d. Dual Coding
Take notes alongside visual representations
Use visual diagrams instead of columns of text

e. Concrete Examples
Come up with or research concrete examples for complex, abstract subjects
I.e, instead of the theory of classical conditioning, refer to Pavlov's dogs

5. **Take a moment to write about your major takeaways from the reading.** Be prepared to share your takeaways in a small group discussion at the beginning of the next period.

- spaced repetition > cramming
- retrieval practice > rereading notes
- tools to use:
    - Interleaving - switch subjects frequently
    - Elaboration - find more meaning
    - Mnemonic Devices
    - Dual Coding - Text + Visual info
    - Concrete examples, not just abstract concepts

Save the space below for notes from your small group discussion…

- as gr 12s, we can be arrogant about our learning
- It is imperative to incorporate retrieval practice & spaced repetition
- Since AP Psych deals w/ th our own brains, actions, etc. students can easily relate it to their lives